

Recognition Without Recall: Diagnosing and Repairing Biologically-Inspired Memory Consolidation in Transformers

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Abstract

This work is relevant to researchers in continual learning, memory-augmented transformers, and biologically-inspired machine learning. We implement and empirically evaluate a biologically-faithful memory consolidation architecture for Large Language Models, drawing on the synaptic tagging-and-capture hypothesis, the Complementary Learning Systems framework, and hippocampal-neocortical replay during sleep. Our system, SLEEP, augments pretrained transformers with five interacting components: prediction-error tagging, direct-write key-value (KV) memory, protein-resource-allocation prioritization, generative replay, and LoRA-based consolidation under explicit safety constraints. The paper proceeds in two movements. The first is a diagnosis: across 200 single-exposure synthetic facts on four models spanning three families (Qwen2.5-7B and 1.5B, Llama-3.1-8B, Mistral-7B), five seeds each, the architecture as originally specified produces recognition without recall—memories that bias multiple-choice judgement but never surface in free-form generation, a floor of DRA ≈ 0.006 against in-context ceilings up to 0.99—while its internal surprise-based validation falsely confirms 93–100% of consolidations and a retrieval-aware warm-up repair fails in sixteen runs. The second movement locates and removes the causes. Guided by the knowledge-editing literature, we show the consolidation failure decomposes into three stacked errors: the adapter was in the wrong substrate (attention projections in the top third of layers, chosen by biological analogy, rather than the mid-stack MLPs where causal tracing places factual associations), it was trained on single wordings (producing memorization without extraction), and the safety machinery’s hard clip erased what training wrote. Correcting all three—mid-stack MLP placement, paraphrase-diverse replay, distillation from an in-context teacher, and a moderate clip—raises single-cycle recall from the 0.006 floor to 0.75 on Mistral-7B (five seeds) with background damage near the acceptability bar, and to 17–125 $\times$ the floor on every model tested. In the multi-cycle regime the machinery was designed for, the repaired system beats naive LoRA on cumulative recall on all four models on both seeds (3 $\times$ on Qwen2.5-7B), beats a matched EWC-only baseline (0.174 vs. 0.056), and prevents the catastrophic damage blow-up that destroys the unconstrained baseline at 1.5B scale (BCP 2.4 vs. 24.5). All validation-phase gates and predictions were pre-registered before the experiments ran. The recognition–recall dissociation, the proxy-validation failure, and the horizon-and-seed instability of unconstrained baselines stand as diagnostic contributions; the localisation repair converts them from a negative result into a working, baseline-controlled system with named residual gaps.

1 Introduction

Large language models exhibit a form of architectural anterograde amnesia: pretrained weights are fixed at deployment, and any new information acquired during inference exists only within the context window, vanishing once the conversation ends. Retrieval-augmented generation (RAG) treats this as plumbing—append a database lookup to every query—rather than as a representational gap to close [Lewis et al., 2020].

Biological memory faces an analogous problem and solves it through a structured pipeline: surprising experiences are flagged at the synapse by prediction-error-driven tags, those tags compete for a limited pool of plasticity-related proteins (PRPs), and the resulting episodic traces are gradually transferred from hippocampus to neocortex through replay during sleep. We review this literature in Section 2 and Section 3;

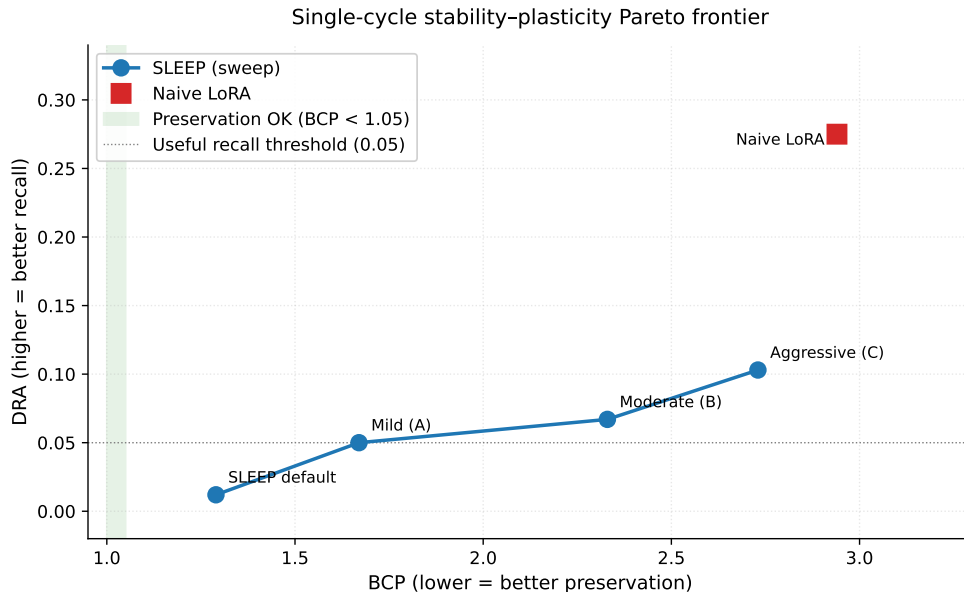


Figure 1: Single-cycle stability-plasticity Pareto frontier of the *original* architecture (attention-top placement, single-wording replay). SLEEP configurations sweep from default (most conservative) to aggressive; naive LoRA marks the unconstrained extreme. The green band shows the preservation-acceptable region ($\text{BCP} < 1.05$); the dotted line shows a useful-recall threshold ($\text{DRA} > 0.05$). No setting on this frontier reaches the upper-left intersection—and Section 7.8 shows why: the frontier itself is an artefact of consolidating into the wrong substrate. The localisation-corrected pipeline of Section 7.10 reaches $\text{DRA} = 0.19$ at $\text{BCP} = 0.88$ on this model (five seeds), off this curve’s scale.

here we note only that the biological result is continuous, selective, single-exposure learning that does not catastrophically overwrite existing knowledge.

We propose SLEEP, a system that implements all five of these mechanisms—tagging, key-value (KV) memory, protein-resource allocation, sleep replay, and dual-weight consolidation—and evaluate each component in isolation across four models from three families. Following a pre-registered formalization in which 36 design questions were resolved before any code was written, we instrument every stage of the pipeline and identify exactly where information is preserved, transformed, or lost. This paper tells the story in the order the evidence forced it on us, in two movements.

The first movement is a diagnosis. The encoding components work: direct-write KV memory injection produces a clear recognition signal on multiple-choice evaluation. The consolidation components, as originally specified, do not: no configuration achieves a Delayed Recall Accuracy (DRA) above 0.05 while holding Base Capability Preservation (BCP) below 1.05, and the recall floor sits at $\text{DRA} \approx 0.006$ on every one of four models across three families, five seeds each—from models that follow the same facts in context at up to 0.99. The failure survives a principled repair attempt (a retrieval-aware warm-up; sixteen runs, no movement), the system’s internal validation certifies consolidations at a 93–100% false-confirmation rate everywhere, and long-horizon comparisons of the unconstrained baseline prove unstable in both horizon and seed.

The second movement locates the causes and removes them. The knowledge-editing literature places factual associations not where biological analogy placed our adapter (attention projections, top third of layers) but in mid-stack MLP modules. Guided by this, a five-arm isolation experiment shows the consolidation failure decomposes into three stacked causes: wrong substrate, single-wording training (which produces string memorization rather than extractable knowledge), and safety machinery whose hard clip erased the write. Correcting all three yields a repaired pipeline—mid-stack MLP placement, paraphrase-diverse replay, distillation from an in-context teacher, moderate clipping—that we then validate under pre-registered gates across the full matrix: five seeds per model single-cycle, ten-cycle continual learning against naive LoRA on all four models, and a matched EWC-only baseline. Single-cycle recall reaches 0.750 ± 0.033 on Mistral-

7B—125 times the old floor—and exceeds it by at least $17\times$ everywhere. In the multi-cycle design regime the repaired system beats naive LoRA on cumulative recall on all four models on both seeds, beats EWC-only three-fold on the anchor model, and prevents the damage catastrophe that seed-randomly destroys the unconstrained baseline at small scale.

These findings yield the following contributions.

Contributions:

- (C1) LoRA fast weights cannot encode at single exposure: $\alpha_{\text{fast}} = 10^{-4}$ falls below the bfloat16 precision floor, and float32 testing shows the default magnitude is separately under-scaled at 7B.
- (C2) Direct-write KV memory injection produces recognition but not generative recall—a transformer analogue of the cognitive recognition–recall dissociation. The recall floor replicates at $\text{DRA} \approx 0.006\text{--}0.007$ on four models across three families, five seeds each, while the same models follow in-context facts at up to 0.99; a retrieval-aware warm-up moves recognition but never recall in sixteen runs.
- (C3) Internal validation passes while external recall fails: across all four models, 93–100% of candidates that pass surprise-reduction fail a direct recall gate—a self-grading discrepancy not unique to SLEEP.
- (C4) Fixed-horizon, single-run comparisons of *unconstrained* continual-learning baselines are unreliable: the damage trajectory of naive fine-tuning varies by up to three orders of magnitude across seeds of identical configurations, flipping fixed-horizon orderings seed to seed.
- (C5) The consolidation failure is causally decomposed into three stacked errors—substrate (attention-top vs. mid-stack MLP), surface diversity (single wording vs. paraphrases), and safety clipping—each isolated experimentally. Placement alone halves background damage in 8 of 8 comparisons; paraphrase diversity trades cloze accuracy for free-form recall (memorization $\rightarrow$ extraction); the designed clip nullifies the recipe (0.007) that a moderate clip preserves (0.21 ± 0.02).
- (C6) The repaired pipeline generalises across families and scales under pre-registered gates: single-cycle trained-subset DRA of 0.750 ± 0.033 (Mistral-7B), 0.512 ± 0.048 (Llama-3.1-8B), 0.189 ± 0.031 (Qwen2.5-7B), and 0.105 ± 0.020 (Qwen2.5-1.5B), five seeds each—17–125 $\times$ the original floor, strongest off the development model.
- (C7) In the ten-cycle regime the machinery was designed for, the repaired system beats naive LoRA on cumulative recall on 4 of 4 models (both seeds; $3\times$ on Qwen2.5-7B) and beats a matched EWC-only baseline (0.174 vs. 0.056): regularization prevents forgetting but does not create extraction. Right-sized safety machinery is load-bearing at small scale, holding BCP near 2.4 where naive LoRA blows up to 24.5, and reduces across-seed variance by an order of magnitude.

The claim structure is deliberately scoped. The diagnostic findings (C1–C4) replicate across every model we can test at this scale. The repair (C5–C7) is validated across three families and two scales with pre-registered gates, with one family flagged: Llama-3.1-8B gains recall like the others but no tested clip setting yet holds its background damage inside our acceptability bar, so we report it as *not yet tuned* rather than solved. We release the implementation, formalization, pre-registrations, experimental logs, and all 35 validation-phase runs.

2 Related Work

We organize related work by methodology rather than chronologically, highlighting how SLEEP relates to each line.

Continual learning via regularization: Kirkpatrick et al. introduced Elastic Weight Consolidation (EWC), which adds a Fisher-information-weighted quadratic penalty to anchor important parameters [Kirkpatrick et al., 2017]. Li and Hoiem proposed Learning Without Forgetting, which uses knowledge-distillation losses against the pre-adaptation model [Li and Hoiem, 2017]. Both require choosing parameters to protect, and both implicitly trade plasticity for stability. SLEEP inherits EWC as one of its safety mechanisms but compounds it with hard clipping, interleaving, and validation rollback; our results measure this compound at

three calibrations: as originally designed it erases consolidation outright (Section 7.9), fully removed it permits catastrophic damage at small scale, and at a moderate setting it becomes load-bearing (Section 7.11). A matched EWC-only arm in the same experiment shows the penalty preserves capability without creating recall.

Continual learning via replay: Shin et al. proposed deep generative replay, which avoids storing raw old data by training a generator to produce old-task samples [Shin et al., 2017]. Rebuffi et al. introduced iCaRL, which maintains a small exemplar buffer and trains with rehearsal [Rebuffi et al., 2017]. Chaudhry et al. showed that even very small replay buffers suffice for many continual-learning settings [Chaudhry et al., 2019]. SLEEP’s sleep cycle is conceptually closest to generative replay—a compressed sample of the experience drives consolidation—but our results in Section 7.3 and Section 7.4 reveal a specific failure mode: in transformer language models, the generative step does not faithfully reproduce source content even when the source is available in attention via a memory bank, because the autoregressive prior dominates sampling.

Memory-augmented transformers: Wu et al. proposed Memorizing Transformers, which augment standard self-attention with a k -nearest-neighbour lookup over a database of past key-value (KV) pairs and fine-tune the model to use that bank during prediction [Wu et al., 2022]. Our KV memory bank borrows the substrate from this line. The crucial difference is that Memorizing Transformers train the model on the memory-augmented objective, whereas we begin from a frozen pretrained model and ask whether the bank can drive learning without retrieval-aware fine-tuning; the recognition-without-recall result in Section 7.2 is, in part, an explanation of why such fine-tuning is necessary, and Section 7.7 reports what a lightweight version of it does and does not recover. Schmidhuber introduced fast-weight programmers, which predate the transformer but express the same idea in a different substrate—memory implemented as outer-product weight updates [Schmidhuber, 1992].

Knowledge editing and localisation: A distinct line asks precisely the question our consolidation step struggles with: how does one write a single fact into a transformer’s weights without breaking the model? Geva et al. showed that feed-forward layers behave as key–value memories storing factual associations [Geva et al., 2021]. Meng et al. used causal tracing to localise those associations to *mid-stack MLP* modules and edit them directly (ROME), then scaled the technique to thousands of simultaneous edits (MEMIT) [Meng et al., 2022, 2023]. Two implications bear directly on this paper. First, the editing literature’s target — mid-stack MLPs — differs from our original adapter placement (attention V/O projections in the top third of layers, chosen by biological analogy), which motivates the localisation experiments of Section 7.8. Second, editing methods demonstrate that single-fact weight writes at preserved general capability are achievable, which reframes our consolidation failure as a question of *where and how* the write happens rather than an inherent stability–plasticity impossibility. Sections 7.8–7.11 confirm this reframing causally: moving the consolidation adapter to the editing literature’s target, and training it on paraphrase diversity, converts the failure into a working system.

Gated memory against frozen backbones: A recent line keeps the backbone frozen and learns only a small component that governs how memory is used. Wang et al. equip a transformer with a learnable write-gate and neural router for self-updatable memory [Wang et al., 2024], and G-MemLLM pairs a frozen backbone with a gated latent memory bank, reporting a 13.3% accuracy gain on ZsRE for Llama 3.1 [G-MemLLM Authors, 2026]. Fountas et al. segment the KV cache by *surprise* for infinite-context retrieval—a mechanism arrived at independently of, and closely resembling, our tagging layer [Fountas et al., 2025]. These results motivate the memory gate of Section 5.6: they establish that a small trainable component can, in principle, teach a frozen model to use external memory. Our contribution to this line is a negative one—we report the scale at which that principle fails to transfer into free-form recall inside a consolidation pipeline.

Retrieval-augmented generation: Lewis et al. attach a learned retriever and an external corpus to a generator at inference time [Lewis et al., 2020]. RAG and SLEEP differ in design intent: RAG keeps memory external and re-queries it for every generation, whereas SLEEP attempts to internalize memory into weights

so that the model becomes the database. Our results suggest that retrieval-from-attention (the recognition signal) is achievable without retraining but retrieval-into-generation (free-form recall) is not, so the SLEEP–RAG distinction collapses partly: the systems we study still require an attention-side mechanism to make encoded knowledge accessible during generation.

Biologically-inspired machine learning: McClelland et al. proposed Complementary Learning Systems (CLS) theory, which has motivated dual-system architectures in machine learning for two decades [McClelland et al., 1995]; most implementations, however, stop at the conceptual level—two networks with different learning rates—without instantiating the full biological pipeline of tagging, PRP allocation, and sleep replay. Frey and Morris characterized synaptic tagging and capture (STC) [Frey and Morris, 1997], which Redondo and Morris later extended [Redondo and Morris, 2011]; Wilson and McNaughton, and Diekelmann and Born, characterized hippocampal–neocortical replay during sleep [Wilson and McNaughton, 1994, Diekelmann and Born, 2010]. SLEEP is, to our knowledge, the most faithful instantiation of these mechanisms attempted in a transformer language model. Our results characterize precisely which biological mechanisms transfer (tagging, attention-based recognition, resource allocation) and which fail in the form proposed (generative replay-driven consolidation under tight safety).

3 Background

We summarize the biological and machine-learning foundations on which SLEEP is built.

Synaptic tagging and capture: The STC hypothesis posits that weak synaptic stimulation creates short-lived molecular “tags” that can later capture PRPs synthesized in response to a separate strong stimulus, converting transient potentiation into durable change [Frey and Morris, 1997, Redondo and Morris, 2011]. STC explains how the brain implements selective consolidation: tags mark candidates, PRPs are the limited resource, and competition among tags determines which traces become permanent. Nader et al. showed that previously consolidated traces become temporarily labile when re-accessed, allowing revision [Nader et al., 2000].

Complementary learning systems: CLS theory proposes two memory subsystems with distinct learning rates and storage formats: a fast hippocampal system that stores arbitrary episodic associations through one-shot binding, and a slow neocortical system that gradually accumulates statistical regularities [McClelland et al., 1995]. Sleep replay bridges the two systems through hippocampal reactivation of recent experiences, training the neocortex while interleaving consolidated knowledge to prevent overwriting [Wilson and McNaughton, 1994, Diekelmann and Born, 2010].

Parameter-efficient fine-tuning: Hu et al. introduced LoRA, which factors weight updates as $\Delta W = BA$ with $B \in \mathbb{R}^{d \times r}$, $A \in \mathbb{R}^{r \times d}$, and $r \ll d$, training only A and B while freezing the base model [Hu et al., 2022]. LoRA enables targeted weight updates without full fine-tuning, making it the natural substrate for the slow consolidation system in our architecture.

Memory-augmented attention: Standard transformer attention computes scores between learned queries and keys derived from the current context window [Vaswani et al., 2017]. As reviewed in Section 2, memory-augmented variants extend this with a lookup over stored KV pairs from past contexts; both Memorizing Transformers and earlier fast-weight programmers rely on retrieval-aware fine-tuning to teach the model to use that memory effectively [Wu et al., 2022, Schmidhuber, 1992]. We adopt the KV-memory substrate but place it within a complete biologically-grounded consolidation pipeline rather than treating it as a standalone retrieval mechanism.

Continual learning mechanisms: Our consolidation step inherits two mechanisms reviewed in Section 2: the Fisher-weighted EWC penalty against catastrophic forgetting [Kirkpatrick et al., 2017], and generative replay, which interleaves synthesized old-task samples with new-task training [Shin et al., 2017, Rebuffi et al., 2017].

4 Problem Statement and Research Questions

The mechanisms reviewed in Section 2 and Section 3 are biological. The open question this paper addresses is whether they can be implemented faithfully inside a frozen pretrained transformer, and which of them survive the transfer. We decompose this into four research questions:

- (RQ1) Can a single inference pass deposit retrievable knowledge into a fast-weight store?
- (RQ2) Does information sufficient for recognition also support free-form autoregressive recall?
- (RQ3) Can a sleep-style consolidation step move that knowledge into permanent weights without degrading base capability?
- (RQ4) Do safety mechanisms that appear over-engineered on a single cycle earn their cost across many cycles?
- (RQ5) When consolidation fails, is the failure a structural limit of the stability–plasticity tradeoff, or a localisation error—and if the latter, does correcting the localisation produce a system that beats standard baselines in its design regime?

We resolved 36 such design questions in a pre-registered formalization before any code was written; the five amendments that empirical work forced are documented in Annexure A. The remainder of the paper answers RQ1–RQ4 in turn (Sections 7.1–7.6), then RQ5 (Sections 7.8–7.11).

5 The SLEEP Architecture

SLEEP augments a frozen base model W_{slow} with three additions:

- (i) A tagging layer that flags surprising spans during inference.
- (ii) A KV memory bank (W_{fast}) into which tagged spans are written one-shot.
- (iii) A consolidation adapter (W_{cons} , a LoRA module) that is updated during periodic sleep phases via replay-driven gradient descent under safety constraints.

A PRP allocator prioritizes which tags get consolidated. Figure 2 summarizes the system. Throughout, labels of the form $Qx.y$ refer to the correspondingly numbered design question in the pre-registered formalization document (Annexure A).

5.1 Tagging Layer

For each input token t at position i , the layer computes the model’s per-token cross-entropy surprise $e_t = -\log p(x_t \mid x_{<t})$. An adaptive z-score thresholder maintains running mean μ and variance σ^2 via an exponential moving average ($\beta = 0.99$) and flags positions where $(e_t - \mu)/\sigma > \kappa$ (default $\kappa = 1.5$). Adjacent flagged positions within $g = 3$ tokens are merged into spans; spans shorter than 4 tokens are discarded. Each surviving span produces a *tag* carrying a key vector $k = \text{Proj}(\bar{h})$ derived from the mean hidden state, an initial strength $s_0 \propto \tanh(E_{\text{span}}/\alpha)$, the source identifier, and the token range. Tags decay exponentially in time and are reinforced when accessed by similar future inputs (Q1.1–1.6).

5.2 KV Memory (W_{fast})

A central design correction in this work is that *tags are pointers, not storage*. The original formalization stored only the K and V projections of the tagged tokens themselves; in practice this captured isolated entity mentions (“Nimbus Holdings”) without their factual contents (revenue figures, dates). We replace this with *episode-level* storage: when a tag fires, we extract pre-RoPE keys and raw values for the full sentence containing the tag from each adapted attention layer. Concretely, for adapted layer ℓ , the memory key and value are computed by two separate projections of the layer-normed episode hidden state:

$$K_{\text{mem}}^{(\ell)} = W_K^{(\ell)} \cdot \text{LayerNorm}_\ell \left(h_{\text{episode}}^{(\ell)} \right), \quad (1)$$

$$V_{\text{mem}}^{(\ell)} = W_V^{(\ell)} \cdot \text{LayerNorm}_\ell \left(h_{\text{episode}}^{(\ell)} \right), \quad (2)$$

SLEEP architecture: wake (left/middle) and sleep (right)

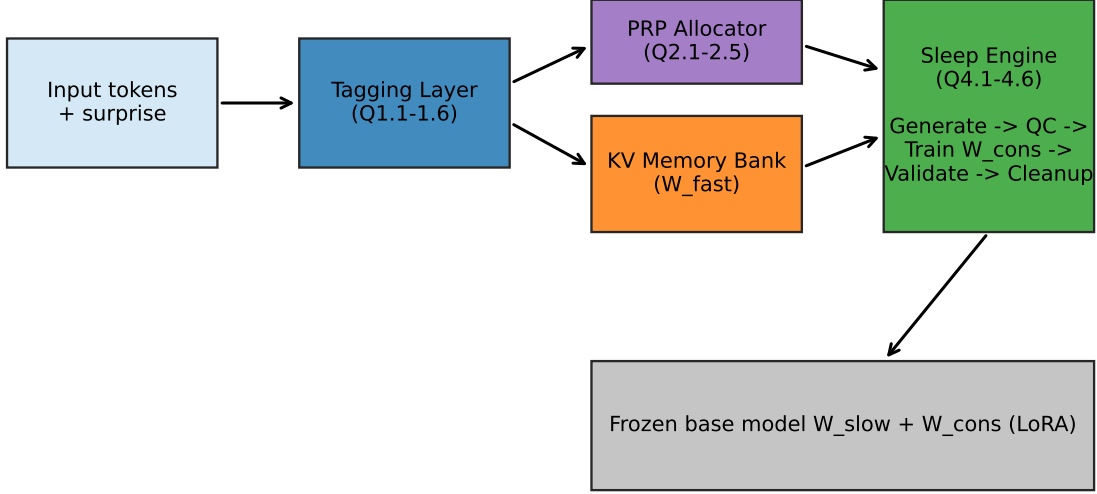


Figure 2: The SLEEP architecture showing wake phase (left) and sleep phase (right). Wake: input tokens are processed by the tagging layer; tags allocate against the PRP budget and write episode-level K/V into the memory bank. Sleep: bank-conditioned generative replay produces synthetic training samples that drive gradient updates to W_{cons} under EWC and hard-clipping safety constraints, after which the bank is cleared.

where $h_{\text{episode}}^{(\ell)}$ is the input hidden state at layer ℓ across the full episode, captured via `output_hidden_states=True`. Crucially, the layer-norm is applied before the projections to match what the model computes during its own attention forward pass; without this correction, extracted K/V differ from in-attention K/V by exactly one normalization, producing silently wrong attention scores.

At inference, stored K/V are injected as a prefix at *negative position IDs* $[-n_{\text{mem}}, -1]$ in each adapted layer’s attention. We apply Rotary Position Embedding (RoPE) to memory keys at read time [Su et al., 2024]; because RoPE is a relative-position rotation, the memory at negative positions appears to current queries as “recent prefix tokens,” a regime the model was trained to handle. Following Wu et al., per-query top- k retrieval gating restricts attention to the k memory positions whose Q·K inner product is highest, implemented through a $-\infty$ additive mask on excluded positions in the model’s existing attention machinery [Wu et al., 2022].

The bank is per-tag indexed and supports per-tag eviction. Capacity is bounded by total stored tokens; the capacity policy is delegated to the PRP allocator.

5.3 PRP Allocation

A composite score combines four signals (Q2.1):

$$\text{score}(\tau) = w_1 \cdot E_{\text{span}}^{\text{norm}} + w_2 \cdot \text{access}^{\text{norm}} + w_3 \cdot \text{xref} + w_4 \cdot e^{-\Delta t / \tau_r}, \quad (3)$$

with weights (0.35, 0.30, 0.15, 0.20). Cross-reference density `xref` measures how many other tags have similar keys (cosine similarity > 0.5). PRP allocation is competitive: only tags whose score exceeds an adaptive threshold θ_{prp} receive an allocation token $p = 1$, and the budget $B = c_{\text{prp}} \cdot N_{\text{params}}^{(B)}$ caps the total. Allocated tags become candidates for sleep consolidation.

5.4 Sleep Engine

A sleep cycle proceeds in five phases, with KV memory injection enabled or disabled per phase to enforce the intended semantics:

1. **Generate** (KV active). For each candidate tag, produce a replay sample by autoregressively sampling from the model conditioned on the early tokens of the source episode. With KV memory active, the model can attend to the original experience while compressing it.
2. **Quality check**. Replays are retained if (a) cosine similarity to the source > 0.5 and (b) mean per-token surprise exceeds a calibration threshold μ_{surprise} .
3. **Train** (KV disabled). W_{cons} is updated by gradient descent on the accepted replays interleaved 1:1 with self-generated samples drawn from the unadapted base model (the “old knowledge” arm). Training disables KV injection so that W_{cons} learns to encode the replay content into weights rather than relying on the bank as a crutch.
4. **Perplexity check** (optional, KV disabled). On benchmark data, post-training perplexity is compared to pre-training perplexity. If degradation exceeds $\epsilon_{\text{degrade}}$, W_{cons} is rolled back.
5. **Validate & cleanup** (KV disabled). For each candidate tag, the post-training model’s surprise on the original span is compared to the pre-training surprise; the tag passes if the reduction is at least ϵ_{learn} (default 10%). Passing tags are removed from the buffer; failing tags are penalized; tags that fail repeatedly are evicted permanently.

After successful consolidation, the KV bank is cleared—its content has, in principle, transferred to W_{cons} .

5.5 Safety Machinery

W_{cons} training applies four safety mechanisms simultaneously. Each targets a distinct failure mode of replay-driven gradient updates:

- **EWC penalty**: a Fisher-weighted quadratic regularizer pulls W_{cons} toward its pre-training checkpoint, with strength λ_{EWC} (default 100) [Kirkpatrick et al., 2017].
- **Hard clipping**: after each optimizer step, per-parameter changes are clipped to magnitude δ_{max} (default 0.01) relative to the pre-cycle checkpoint, and gradients are scaled by per-layer plasticity coefficients $\phi_{\ell} \in [\phi_{\text{min}}, 1]$ that encode “hippocampal-like” top-third layers as more plastic.
- **Old-knowledge interleaving**: each training step mixes replay samples with self-generated continuations from the unadapted base model, providing regularization pressure toward preserving base capabilities.
- **Validation rollback**: candidates whose surprise on the original span does not drop by $\epsilon_{\text{learn}} = 0.10$ are rolled back to their pre-cycle parameter values.

The cumulative effect is a strong stability bias. The empirical question is whether this bias is correctly calibrated. The answer turns out to be three-valued: as designed, the clip erases consolidation entirely (Section 7.9); removed, it permits catastrophic damage at small scale; at a moderate setting ($\delta_{\text{max}} = 0.1$) it preserves both the write and the protection (Section 7.11).

5.6 Retrieval-Aware Warm-Up

The five components above constitute SLEEP as originally specified. We add one further component, motivated directly by the empirical finding in Section 7.2: memory that supports recognition does not support generation, because a frozen pretrained model has never been trained to condition its next-token distribution on retrieved K/V. Wu et al. resolve this by training the model on a memory-augmented objective [Wu et al., 2022]; gated-memory architectures reach the same end with a small trainable component against a frozen backbone [G-MemLLM Authors, 2026, Wang et al., 2024]. We adopt the lightest form of this idea that fits inside SLEEP.

Mechanism: Before any continual learning begins, we run a one-time warm-up on a general corpus that contains none of the evaluation facts. For each step we split a passage into a prefix and a continuation, write the prefix’s episode K/V into the memory bank, and train on the continuation with that memory injected. Because the prefix is genuinely informative about the continuation but reaches the model only through memory, lowering the loss requires learning to route memory into generation. The warm-up therefore teaches a *skill*, not a fact.

The trainable component: We introduce a *memory gate*: a per-layer, per-KV-head multiplicative scale $g_h^{(\ell)} > 0$ applied to the memory values before they enter attention,

$$\tilde{V}_{\text{mem}}^{(\ell)} = g^{(\ell)} \odot V_{\text{mem}}^{(\ell)}. \quad (4)$$

Because attention output is linear in V and the softmax weights depend only on $Q \cdot K$, this scales exactly the memory contribution to the residual stream and nothing else. The gate is parameterised as a log-scale initialised at zero, so an untrained gate is exactly the identity and installing it changes no computation. It holds $|\mathcal{L}_{\text{adapted}}| \times n_{\text{kv-heads}}$ parameters—36 for the top third of Qwen2.5-7B. We also test a higher-capacity variant in which the W_{cons} LoRA adapter trains alongside the gate.

Crucially, the gate lives on the attention path rather than inside W_{cons} , so it persists across sleep cycles that overwrite the consolidation adapter. Section 7.7 reports what this buys.

5.7 The Localisation Revision (Pipeline v2)

The components above constitute the architecture whose diagnosis occupies Sections 7.1–7.7. The knowledge-editing literature reviewed in Section 2 motivates a revision of the consolidation step along three axes, which we specify here and isolate experimentally in Section 7.8.

Substrate: mid-stack MLP placement. The original W_{cons} adapts attention V/O projections in the top third of layers—a placement chosen by analogy to hippocampal–neocortical anatomy. Causal-tracing work places factual associations in mid-stack MLP modules [Geva et al., 2021, Meng et al., 2022, 2023]. The revised W_{cons} therefore adapts the MLP up- and down-projections of the middle third of layers. Because KV memory injection has independent, verified requirements (top-third attention), we decouple the two placements: the configuration gains separate `layer_selection` (adapter) and `injection_selection` (bank) parameters, and the memory bank continues to live on top-third attention while consolidation targets mid-stack MLPs.

Surface diversity: paraphrase replay. Training a fact’s single canonical wording teaches the string, not the fact: cloze completion succeeds while question-answering fails. The revised replay strategy trains each selected fact on a deterministic set of at least 20 paraphrases spanning declarative reorderings, clefts, appositives, and—critically—question-answer forms, generated by seeded slot-filling templates so that datasets are bit-reproducible. Replay diversity converts memorization into extraction (Section 7.8).

Objective: distillation from an in-context teacher. The strongest recall any model exhibits is with the fact in context (Table 5). The revised training objective exploits this: for each paraphrase, the *teacher* distribution is the same model with the fact placed in context (adapter disabled), and W_{cons} is trained to match it without the context,

$$\mathcal{L} = T^2 \text{KL}\left(p_{\text{teacher}}^{(T)} \parallel p_{\text{student}}^{(T)}\right) + \alpha_{\text{CE}} \mathcal{L}_{\text{CE}}, \quad (5)$$

with temperature $T = 2$ and $\alpha_{\text{CE}} = 0.5$, KL computed per token over the flattened (position, vocabulary) axes. This is context distillation applied inside a consolidation loop: the model teaches its adapter what it already does correctly when it can see the fact.

Calibration: moderate safety and an external gate. The revised pipeline runs the identical five-phase sleep engine and safety machinery at a re-calibrated operating point ($\delta_{\max} = 0.1$, plasticity scaling on), motivated by the isolation experiment of Section 7.9. Internal validation is replaced by the two-stage gate of Section 7.4: candidates must pass the direct greedy-decoding recall test, not merely surprise reduction, before being counted as consolidated. All knowledge-presence evaluation in the revised pipeline uses greedy decoding, removing sampling temperature as a confound.

6 Experimental Setup

Models: The primary model is Qwen2.5-7B with bfloat16 weights, eager attention, and no quantization [Qwen Team, 2024]: 28 transformer layers, 28 query heads, 4 key-value heads (grouped-query attention), head dimension 128. To test generality we repeat the core protocol on Qwen2.5-1.5B (same family, weaker prior), Llama-3.1-8B, and Mistral-7B (different families, different pretraining corpora). In the diagnosis-era experiments we adapt the top third of layers for both KV memory injection and the W_{cons} LoRA module; in the revised pipeline (Section 5.7) injection stays on top-third attention while W_{cons} moves to mid-stack MLP projections, the two placements decoupled in configuration. All layer sets are derived from each model’s layer count rather than hardcoded. Before any experiment on a new architecture, a verification harness asserts three ground-truth properties of the injection path on the real model—empty-bank identity, populated-bank effect, and disable-restores-baseline—and the architecture is excluded if any check fails (none were).

Dataset: We construct a controlled benchmark of 200 synthetic facts spanning 10 templates: corporate financials, scientific discoveries, city foundings, regulatory protocols, world records, technology architectures, medical trials, sports records, album releases, and geological events. Each fact is a single sentence containing fictional entity names, specific numerical values, and contextual details. Templates are deliberately structured so that the surprising tokens (entity names like “Nimbus Holdings”) cluster at the start of each fact, with factual content (revenue figures, dates, percentages) following. Each fact is paired with a question prompt and a list of three keywords expected in a correct answer. The dataset is generated by deterministic seeded Python code; see Annexure for the generator. For the revised pipeline, each fact additionally carries 22–25 deterministic paraphrases (declarative reorderings, clefts, appositives, and question–answer forms) produced by slot-filling the same seeded templates; the paraphrase extension draws no additional random numbers, so the 200 base facts are bit-identical between the diagnosis-era and revised datasets.

We deliberately use synthetic facts rather than natural documents to isolate single-exposure encoding from prior-knowledge confounds. Real-world facts may have been seen during pretraining, making it impossible to distinguish “the model recalls because it learned” from “the model recalls because it always knew.”

Evaluation metrics: We measure three forms of recall and one preservation metric:

Multiple-choice accuracy (MC): for each fact, the correct keyword and three same-template distractors are presented as options A–D in a randomized order seeded by fact ID. We compute the model’s logit at each option’s first token following “Answer:” and report the fraction where the argmax matches the correct option. We also log $P(\text{correct})$ and $P(\text{correct} \mid \text{wrong})$ via softmax over the four option-letter logits, providing a calibration view sensitive to partial encoding.

Cloze accuracy: the prefix of the fact text up to (but not including) the first keyword is provided as a prompt; the model generates 20 tokens; we score 1 if the keyword appears in the generation, 0 otherwise.

Free-form delayed recall accuracy (DRA): the fact’s question prompt (e.g., “What was Nimbus Holdings’ Q3 revenue?”) is provided; the model generates 50 tokens; we compute the fraction of expected keywords appearing in the generation. The aggregate DRA is the mean across facts. Diagnosis-era experiments sampled at temperature 0.7; all revised-pipeline evaluation uses *greedy decoding*, so that knowledge-presence claims are not confounded by sampling noise (a reviewer-motivated correction; the two decodings agree at the floor, and greedy is the conservative choice for positive claims). Because the PRP allocator selects a subset of facts for consolidation (~ 30 – 50 of 200 per cycle), we additionally report *trained-subset DRA*—recall over the facts the system actually chose to train—so that selection is not silently charged against recall.

Base capability preservation (BCP): we measure perplexity on three held-out control texts (capital of France, water boiling temperature, Python’s history) before and after each modification, and report the

ratio $\text{BCP} = \text{PPL}_{\text{after}} / \text{PPL}_{\text{before}}$. $\text{BCP} \leq 1.05$ is the preservation threshold (no meaningful degradation); $\text{BCP} > 2$ indicates substantial damage.

Baselines: The **base model** is the unmodified Qwen2.5-7B. **Naive LoRA** is a single LoRA adapter (rank 16, $\alpha = 32$, applied to V/O projections of the same top-third layers as W_{cons}) trained directly on the 200 facts via uniform-with-replacement sampling for 100 gradient steps with AdamW (lr 10^{-4} , weight decay 0.01, batch size 32). This matches SLEEP’s compute budget but applies no safety constraints. We add three reference points that bound the problem from opposite directions: **in-context injection** places the gold fact directly in the prompt with no training at all (an upper bound on what the model can do with a fact it can see); **RAG** retrieves the top-3 facts by mean-pooled embedding similarity and prepends them [Lewis et al., 2020]; and **EWC-only** applies the Fisher-weighted penalty to naive LoRA with none of SLEEP’s other machinery, isolating how much of any preservation benefit is attributable to EWC alone. For the multi-cycle validation matrix, the EWC-only arm is *exactly matched* to the naive arm—identical adapter geometry, steps, learning rate, and batch schedule; the only difference is an online-EWC quadratic penalty ($\lambda = 100$, Fisher accumulated per cycle and re-anchored after each)—so any difference between the two arms is attributable to the penalty and nothing else. The arm’s Fisher accumulation and anchoring are verified by ground-truth unit tests before use.

Seeds and variance reporting: Every headline result is run across multiple random seeds, each in an independent process so that model initialization, tagging decisions, replay sampling, and MC option ordering all vary. The single-cycle pipeline and its recall/preservation metrics use five seeds (0–4) *per model* and we report mean $\pm$ standard deviation throughout. The warm-up experiments use two seeds per variant per model (sixteen runs total). The ten-cycle continual-learning protocol runs three seeds on Qwen2.5-7B and two on each other model (nine run-pairs); because its seed variance turns out to be the finding rather than noise, Section 7.6 reports per-run trajectories rather than averages.

Pre-registration of the validation matrix: The revised-pipeline validation (Sections 7.10–7.11) was run as a staged campaign with gates and predictions committed to the repository before execution: a ground-truth verification gate on every architecture; a tuning stage whose advancement criterion (trained-subset $\text{DRA} \geq 0.10$ at $\text{BCP} \leq 1.5$, best recipe per model) was fixed in advance; quantitative predictions for the single-cycle, multi-cycle, and EWC comparisons; and a stop-loss. One extension (a clip ladder for Llama-3.1-8B) was registered mid-campaign with its selection rule stated before its runs executed. Every gate was read as written, including the one Llama fails. The matrix comprises 35 runs executed without a single failure or re-run.

Warm-up protocol: The retrieval-aware warm-up (Section 5.6) trains on 400 passages drawn from WikiText-2, filtered to 80–400 characters and containing none of the evaluation facts. Each step samples a passage, splits it at a point drawn from the middle two quarters, writes the prefix episode into the bank, and takes one AdamW step (lr 10^{-3} , gradient clipping 1.0) on the continuation with memory injected. We report 300-step runs; a 150-step configuration was also tested and is reported in Section 7.7. After warm-up the gate is frozen and the bank cleared, so evaluation begins from an empty memory exactly as in the no-warm-up condition.

Hyperparameters: All non-overridden values track the formalization defaults. We document four hyperparameter overrides applied during the empirical work, each with its empirical justification logged at the time the change was made (see the logbook entries cited in Annexure A): $\alpha_{\text{slow}} = 10^{-4}$ (precision floor under bfloat16; the 10^{-5} default produces zero detectable parameter movement), $\delta_{\text{max}} = 0.01$ (model-scale-dependent; 10^{-3} saturates within 30 steps on 7B), KV top- $k = 64$ (must scale with episode size; episodes are ~ 30 –40 tokens so $k = 16$ retrieves only fragments), and quality-check μ_{surprise} disabled (the formalization’s absolute-surprise threshold suffers from a generator-discriminator gap on self-generated replays). The full hyperparameter list is given in Table 12 of the Annexure.

Compute: Experiments run on single NVIDIA GPUs via RunPod: an RTX 5090 (32GB) for the initial single-cycle work and RTX A6000s (48GB) for the multi-seed, ten-cycle, warm-up, float32, and cross-model

programmes—single precision at 7B and the full four-model matrix both require more than 32GB. Total compute for all experiments reported in this paper is approximately 50 GPU-hours, totaling roughly USD \$40 in cloud cost, of which the localisation-and-validation programme (ladder through full matrix) accounts for about 25 GPU-hours. Each end-to-end run (200 facts, full sleep cycle, complete evaluation suite) takes 8–15 minutes at 7–8B scale. We log every run’s command line, configuration, output, and resulting metrics; these are released alongside the implementation.

Reproducibility: The implementation comprises approximately 4,000 lines of Python across eight modules (tagging, prp, weights, sleep_engine, orchestrator, evaluation, datagen, distill), with 410 unit and integration tests, including ground-truth equality tests on every critical numerical path (KV extraction/injection identity, teacher–student equivalence at adapter zero-initialisation, EWC Fisher accumulation). The 36-question formalization and the empirical amendments derived from it are released as a separate document to enable independent reproduction. The exact RunPod instance image, Python version, and `transformers` library version (5.7.0) are documented; `attn_implementation=eager` is forced because the Scaled Dot-Product Attention (SDPA) path handles 4D additive masks differently and silently degrades the gating mechanism.

7 Results

The results follow the paper’s two movements. The first movement (§7.1–§7.7) is an anatomy of where in the original SLEEP pipeline information is preserved, transformed, or lost: where encoding succeeds (§7.1), why it does not become recall (§7.2), what storage granularity is required (§7.3), where consolidation stalls (§7.4), what the safety parameters buy (§7.5), how the answers change with evaluation horizon (§7.6), and whether the recognition–recall gap yields to a lightweight repair (§7.7). The second movement (§7.8–§7.11) locates the causes of the consolidation failure and validates their removal: a five-arm isolation of substrate and surface diversity (§7.8), the discovery that the safety machinery itself was a third cause (§7.9), the repaired pipeline as an autonomous system (§7.10), and the pre-registered four-model validation matrix (§7.11).

7.1 Encoding: LoRA Fast Weights vs. KV Memory Injection

Proposed: The original SLEEP design specified LoRA adapters as the fast-weight substrate. We find this configuration cannot deposit retrievable knowledge from a single inference pass, and we show that direct-write KV memory injection produces a clear—if limited—encoding signal.

Setup: We compare two W_{fast} substrates on an identical 200-fact wake-only protocol: (i) LoRA W_{fast} updated by gradient descent on each tagged span (the original specification), and (ii) KV memory injection with episode storage and top- k retrieval gating ($k = 16$). For each, we evaluate multiple-choice accuracy and BCP after wake (no sleep cycle).

Results: Table 1 reports the contrast. LoRA W_{fast} at $\alpha_{\text{fast}} = 10^{-4}$ produces no detectable encoding signal: tagged-fact MC accuracy (0.23) is statistically indistinguishable from untagged-fact accuracy (0.24). Increasing the learning rate to 10^{-3} degrades the model (BCP 1.17) without producing recall. We attribute this to the bfloat16 precision floor: at $\alpha_{\text{fast}} = 10^{-4}$, single-step gradient updates are below the dtype’s representable epsilon for typical weight magnitudes.

KV memory injection with retrieval gating produces a +0.16 multiple-choice gap between tagged and untagged facts, with BCP held at 1.08 (Figure 3).

Calibration analysis is more sensitive: the average probability assigned to the correct option, conditional on the model picking the wrong one, is 0.23 for tagged facts versus 0.19 for untagged—suggesting a partial encoding signal that does not quite flip predictions. Without retrieval gating, the KV bank’s contents drown attention on irrelevant queries, destroying base capability (BCP 96.88).

Table 1: W_{fast} substrate comparison on the 200-fact wake-only protocol. KV injection produces a recognition signal that LoRA does not; KV without retrieval gating destroys BCP.

	MC Tagged	MC Untagged	Δ (T-U)	BCP
LoRA W_{fast} ($\alpha = 10^{-4}$)	0.23	0.24	-0.01	0.99
LoRA W_{fast} ($\alpha = 10^{-3}$)	0.24	0.24	0.00	1.17
KV memory (no gating)	0.27	0.16	+0.11	96.88
KV memory ($k = 16$, gated)	0.28	0.12	+0.16	1.08

KV memory injection produces a recognition signal that LoRA does not

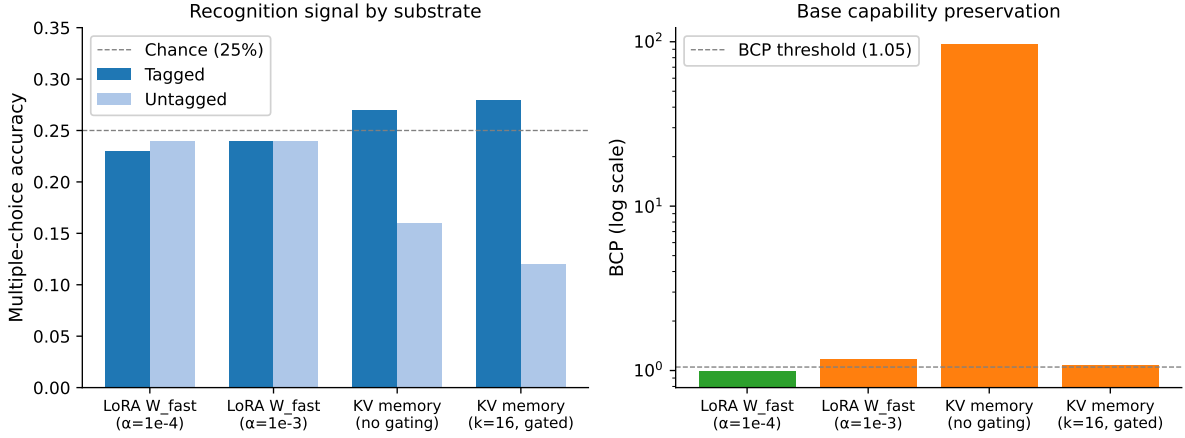


Figure 3: Comparison of W_{fast} substrates on a 200-fact wake-only protocol. Left: tagged-vs-untagged multiple-choice accuracy. Right: BCP (log scale, since the no-gating KV configuration produces $\sim 100\times$ degradation). The KV substrate with retrieval gating is the only configuration that produces a measurable recognition signal (Tagged - Untagged = +0.16) while preserving base capability.

7.2 Recognition Without Recall

The KV memory result above is structurally partial. Tagged-fact *multiple-choice* accuracy exceeds untagged by +0.16, but tagged-fact *cloze* accuracy is 0.00 and tagged-fact *free-form* accuracy is 0.00. The same memories that bias four-way recognition fail to drive autoregressive token generation. We name this dissociation the *recognition-recall gap* and treat it as the paper’s central novel finding: it is not a defect of our specific implementation but, we argue, a structural property of memory-augmented attention over frozen pretrained models.

The cognitive-psychology analogue: The recognition-recall dissociation is a foundational observation in episodic memory research [Morris, 2006]. Recognition (“does this seem familiar?”) can be supported by partial trace reactivation—a sense of fluency or matching—while free recall (“what came next?”) requires reconstructing the trace in sufficient completeness to generate its content. Patients with hippocampal damage often retain residual recognition ability even when free recall is severely impaired. The structural distinction in cognitive psychology is that recognition is a similarity-judgment task (compare the probe to stored representations), while recall is a generation task (produce the stored representation autoregressively from a cue). Our result reproduces this distinction in transformer language models for a different reason at the mechanistic level, but with the same operational consequence.

The mechanistic explanation: At each generation step, attention to memory K/V provides additive perturbations to the pre-softmax logits at every layer. For our $n_{\text{mem}} \approx 5,500$ stored tokens with top- k retrieval gating, the k retained memory positions contribute at most $k/(k + |\text{seq}|)$ of total attention mass,

with the remainder going to the current sequence’s tokens whose keys are RoPE-anchored at the model’s normal training distribution. Memory thus enters the next-token prediction as a small perturbation against a strong prior trained on trillions of tokens. The perturbation is sufficient to flip an argmax in a four-way comparison—all four options share the same prior and the memory tilts the relative scores—but is overwhelmed across an autoregressive generation, where each generated token comes from a categorical distribution over a 150,000-token vocabulary in which the prior dominates. A memory entry can nudge the most likely token of a sentence; it cannot consistently steer 50 generated tokens against the model’s natural continuation distribution.

Why this matters beyond SLEEP: Memorizing Transformers, of Wu et al., avoid exactly this failure mode by retraining the model on memory-augmented losses, teaching the next-token distribution to depend on retrieved K/V [Wu et al., 2022]. Our recognition-without-recall result is—in part—a precise specification of *what* that fine-tuning fixes: the gap between attention-time bias and sampling-time use. For applications that need a frozen pretrained backbone with appended memory, the gap is structural and cannot be closed by hyperparameter tuning. We return to the architectural implications in Section 8.

7.3 Episode Storage: Tags Are Pointers

Proposed: The original formalization stored only tagged-token K/V; we show this captures entity mentions without their factual content, and that storing the full sentence containing the tag is necessary for replay quality.

Diagnostic: We inspect the generated replays produced by the sleep cycle when the bank stores only the tagged span. With seeds like “Nimbus Holdings reported Q”, the model generates fluent financial-news text with fabricated revenue figures (“\$32.6 million”, “up 6.1% from prior year”)—values not in the original fact. The pattern persists across templates: stored entity-name K/V provide an attention bias toward “earnings statement” representations without supplying the specific facts.

When storage is expanded to the full sentence containing the tag (the *episode*), and the seed for replay generation is taken from the start of the episode rather than from the tag location, replay coherence improves: the consolidation rate (validation passes / candidates) rises from 40% to 62%, and BCP improves from 1.06 to 1.008 ($k = 16$, episode storage). The recall metrics, however, do not improve correspondingly—a finding we examine in Section 7.4.

This empirical correction validates the proposal’s own framing: *tags are pointers, not storage*. The implementation must store what the pointer indexes, not the pointer’s surface form.

7.4 The Consolidation Bottleneck

Proposed: Even when generative replay is replaced with the original tokens (eliminating replay-quality as a confound), SLEEP’s safety machinery prevents sufficient W_{cons} updates for free-form recall.

Setup: We add a *replay-strategy* parameter to the sleep engine. With `replay_strategy="original"`, the engine bypasses generation entirely and uses the original fact tokens as replays. All other components—PRP allocation, EWC, hard clipping, validation, rollback, KV-disabled training—run unchanged. This isolates “does W_{cons} training learn from data, given perfect data?”

Result: With original-token replays under default safety parameters ($\delta_{\text{max}} = 0.01$, $\lambda_{\text{EWC}} = 100$, $\alpha_{\text{slow}} = 10^{-4}$, ~ 100 steps), recall remains at floor across five seeds: $\text{DRA} = 0.006 \pm 0.003$ with $\text{BCP} = 1.066 \pm 0.032$. Perfect replay data does not rescue consolidation.

Quantifying the self-grading discrepancy: To measure—rather than argue—the mismatch between internal and external criteria, we add a second validation stage. Stage one is the original surprise-reduction check; stage two is a direct free-form recall test that asks whether the post-training model can produce any expected keyword from the fact’s question prompt under greedy decoding. Table 2 reports both gates across five seeds.

Table 2: Two-stage validation across five seeds (200 facts, original-token replay). The surprise proxy confirms roughly 39 consolidations per run; fewer than one survives a direct recall test.

Quantity	Mean $\pm$ SD	Per-seed values
Passed surprise-reduction (stage 1)	38.8 ± 0.4	39, 39, 39, 39, 38
Passed direct recall gate (stage 2)	0.4 ± 0.5	0, 1, 1, 0, 0
DRA	0.006 ± 0.003	.005, .007, .010, .003, .005
BCP	1.066 ± 0.032	1.11, 1.03, 1.04, 1.09, 1.06

Table 3: The recall floor and self-grading gap across four models, three families (5 seeds each, mean $\pm$ SD). The floor and the proxy–recall mismatch replicate everywhere.

Model	DRA	BCP	Proxy passes	Recall-gate passes
Qwen2.5-7B	0.006 ± 0.003	1.066 ± 0.032	38.8 ± 0.4	0.4 ± 0.5
Qwen2.5-1.5B	0.007 ± 0.002	0.997 ± 0.003	23.8 ± 1.1	0.0 ± 0.0
Llama-3.1-8B	0.006 ± 0.003	0.958 ± 0.009	9.2 ± 2.4	0.6 ± 0.5
Mistral-7B	0.007 ± 0.000	1.001 ± 0.002	4.0 ± 1.0	0.0 ± 0.0

The gap is a factor of roughly one hundred: **$\sim 97\%$ of the consolidations the system certifies internally fail an external recall test**, and the effect is consistent on every seed. *Internal validation criteria can pass while external recall fails.* We argue this self-grading discrepancy is a methodological concern broader than SLEEP: a system that grades itself on a proxy of its own behavior provides no protection against proxy–target mismatch, and the magnitude here suggests the mismatch can be near-total rather than marginal.

Cross-family and cross-scale replication: Because both the recall floor and the self-grading gap could in principle be artefacts of one model’s pretraining, we repeat the five-seed protocol on three further models spanning two additional configurations of the same family and two other families. Table 3 shows the outcome: the free-form recall floor sits at $\text{DRA} \approx 0.006\text{--}0.007$ on every model, and the direct recall gate confirms at most a fraction of one consolidation per run everywhere, against 4–39 proxy confirmations. The number of *candidates* varies substantially across families—Mistral’s surprise statistics allocate a tenth as many as Qwen—but the false-confirmation structure is invariant. Notably, the same models follow in-context facts well (Table 5), so the parametric recall floor is not a comprehension failure.

7.5 The Stability–Plasticity Pareto Frontier

Proposed: Sweeping the four safety knobs produces a smooth Pareto curve with no operating point achieving both meaningful recall and preservation on the single-cycle benchmark.

Setup: Using original-token replay (clean signal), we sweep four settings of $(\delta_{\max}, \lambda_{\text{EWC}}, \alpha_{\text{slow}}, \text{steps_per_memory})$ from SLEEP defaults to near-naive-LoRA. We compare against naive LoRA fine-tuning at matched compute. Table 4 reports the curve.

The curve (Figure 1, also shown on page 1 as the paper’s central result) is monotonic: relaxing safety raises DRA and raises BCP simultaneously. There is no knee; no inflection point where DRA jumps without BCP cost. On a single-cycle 200-fact benchmark, the SLEEP architecture’s value proposition—“a small recall sacrifice in exchange for guaranteed preservation”—does not hold: every SLEEP point pays more BCP per percentage point of recall than naive LoRA does at its single endpoint.

Reference points: Table 5 places these numbers against three baselines that bound the problem. Supplying the fact in context yields $\text{DRA} = 0.760$ at no preservation cost, establishing that the model reads and uses a fact it can see; RAG’s shortfall against that ceiling (0.213) is therefore attributable to retrieval rather than

Table 4: Single-cycle Pareto frontier on the 200-fact benchmark. No SLEEP setting achieves $\text{DRA} > 0.05$ with $\text{BCP} < 1.05$. Naive LoRA achieves higher DRA at unacceptable BCP cost.

Setting	$\delta_{\max}$	λ_{EWC}	α_{slow}	steps/mem	DRA	BCP
SLEEP default	0.01	100	10^{-4}	5	0.012	1.29
Mild relax (A)	0.02	50	10^{-4}	4	0.050	1.67
Moderate (B)	0.05	10	3×10^{-4}	4	0.067	2.33
Aggressive (C)	0.10	1	5×10^{-4}	6	0.103	2.73
Naive LoRA	∞	0	10^{-4}	—	0.275	2.94

Table 5: Baseline reference points across models (100 facts, single seed each). In-context injection bounds the task from above everywhere; on 7B-class models EWC-only achieves a better recall-per-preservation ratio than any full SLEEP configuration, though the effect weakens at 1.5B.

	Qwen-7B	Qwen-1.5B	Llama-8B	Mistral-7B
In-context DRA	0.760	0.700	0.987	0.927
RAG DRA	0.213	0.460	0.520	0.563
EWC-only DRA (BCP)	0.070 (1.12)	0.033 (1.43)	0.073 (1.47)	0.157 (2.53)

to comprehension. Most pointedly, EWC applied alone—without hard clipping, interleaving, validation rollback, tagging, or sleep scheduling—reaches $\text{DRA} = 0.070$ at $\text{BCP} = 1.116$, a better recall-per-preservation ratio than any full SLEEP configuration in Table 4.

Single-cycle interpretation: A natural reading of these single-cycle results is that SLEEP’s safety machinery is over-engineered. The mechanisms (EWC, hard clipping, interleaving, validation rollback) collectively dampen learning more than they protect base capability, and the EWC-only baseline suggests that the marginal machinery beyond the Fisher penalty is not paying for itself. On a one-shot benchmark, that is a true description of what they do. Section 7.6 tests whether it remains true as the horizon lengthens.

7.6 Evaluation Horizon and Seed: The Long-Run Comparison Is Not Stable

Proposed: SLEEP’s safety mechanisms are designed for repeated-cycle continual learning rather than single-cycle benchmarks. We therefore evaluate in that regime—and find that the comparison against naive fine-tuning is not stable in the number of cycles, which changes what can be claimed.

Setup: We split 200 facts into 10 disjoint batches of 20. Across ten sequential cycles the model sees one batch per cycle. After each cycle we measure (i) DRA on the cumulative facts seen so far (the forgetting test), (ii) DRA on the current cycle’s facts (the learning test), and (iii) BCP on controls (the preservation test). We compare **SLEEP-A** (Setting A from the Pareto sweep) against **Naive LoRA** (200 steps per cycle on that cycle’s batch). Both arms accumulate weight updates across cycles—each cycle starts from the previous cycle’s adapter state. We run this protocol with three seeds on Qwen2.5-7B and two seeds on each of Qwen2.5-1.5B, Llama-3.1-8B, and Mistral-7B: nine SLEEP runs and nine naive runs in total.

Result—a universal shape with a lottery for its level: Table 6 reports all nine run-pairs. Three regularities hold in every run. First, the *step-then-plateau* signature: each SLEEP run holds BCP near 1 for its first cycles, then jumps discretely and stays flat—within a few percent—for every remaining cycle (9/9 runs). Second, naive LoRA’s cumulative recall exceeds SLEEP’s in 8 of 9 pairs, typically by 2–20 $\times$. Third, no run of either arm ever satisfies $\text{BCP} < 1.05$ beyond cycle two. What does *not* replicate is the plateau level: across seeds of the *same configuration* it spans 2.7–10.2 on Qwen-7B, 16–230 on Qwen-1.5B, and 4.8–1531 on Llama—three orders of magnitude on identical hyperparameters. Which arm “wins” preservation at cycle 10 accordingly flips seed to seed (SLEEP wins 4 of 9 pairs, naive 5 of 9).

Table 6: Ten-cycle continual learning across four models (cycle-10 endpoints). “Plateau from” gives the cycle at which the SLEEP arm’s BCP stops changing. Every SLEEP run shows a step-then-plateau trajectory; the plateau *level* varies by orders of magnitude across seeds of the same configuration.

Model	Seed	SLEEP BCP@10 (plateau from)	Naive BCP@10	SLEEP DRA	Naive DRA
Qwen-7B	0	4.53 (c4)	3.36	0.050	0.085
Qwen-7B	1	10.20 (c3)	6.42	0.032	0.080
Qwen-7B	2	2.71 (c5)	5.44	0.057	0.078
Qwen-1.5B	0	15.96 (c3)	21.01	0.045	0.035
Qwen-1.5B	1	230.1 (c5)	45.83	0.012	0.048
Llama-8B	0	4.83 (c3)	6.09	0.032	0.083
Llama-8B	1	1530.9 (c3)	7.11	0.045	0.075
Mistral-7B	0	3.22 (c2)	2.90	0.008	0.192
Mistral-7B	1	1.64 (c2)	3.72	0.003	0.160

Multi-cycle: SLEEP’s preservation advantage is transient, not architectural

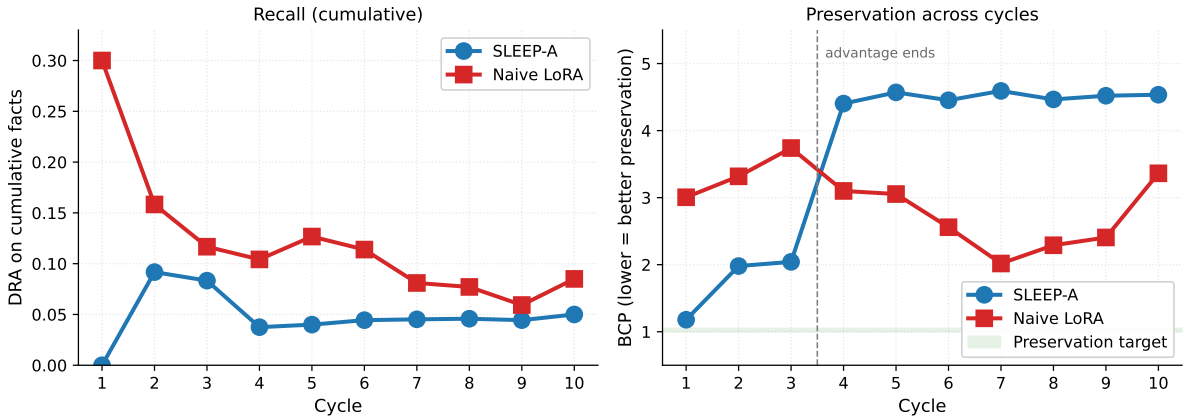


Figure 4: Ten-cycle continual learning on disjoint batches (Qwen2.5-7B, seed 0 shown; Table 6 reports all nine run-pairs). Left: cumulative DRA. Right: BCP with the crossover marked. SLEEP’s preservation advantage holds for three cycles, after which its BCP steps to a plateau while naive LoRA fluctuates.

Interpretation—horizon and variance jointly decide the ordering: Three successively larger experiments gave three different headlines: a single cycle favours naive LoRA; three cycles favour SLEEP (the ordering we ourselves reported from the three-cycle experiment); ten cycles across seeds shows the ordering is not stable at all. The seed-stable claims are structural, not comparative: SLEEP’s clipping mechanism caps damage at a plateau whose *existence* is guaranteed but whose *level* is effectively a lottery over the optimisation path—consistent with $\delta_{\max}$ saturation geometry, where the clip freezes whatever damage has accumulated when saturation is reached—and the unconstrained baseline buys its consistently higher recall with wildly fluctuating damage. We conclude that **single-run, fixed-horizon comparisons of continual-learning architectures are unreliable at both of the scales this field typically reports**, and recommend that such comparisons state a horizon sweep and a seed spread or not be made.

What this result does *not* show: It does not show SLEEP’s safety machinery is useless—in 4 of 9 pairs it produced the better cycle-10 preservation, and its early-cycle protection is real everywhere. It shows the machinery does not deliver a *reliable* advantage, and that the aspirational target—BCP near 1 across many cycles while DRA accumulates—was met by no run of either arm. Section 7.11 revisits this protocol with the repaired pipeline and finds that the seed lottery was a property of the *choked* configuration: once consolidation writes to the correct substrate under a moderate clip, the constrained arm’s trajectories

Table 7: Retrieval-aware warm-up across four models (two seeds per variant per model; Δ DRA ranges span seeds). Recall moves by at most ± 0.008 anywhere. The gate-only variant is harmless but inert; the +LoRA variant learns its objective yet damages base capability by a family-dependent, sometimes catastrophic amount.

Model	Variant	Δ DRA (seeds)	BCP after (seeds)
Qwen-7B	gate only	+0.002, +0.002	~ 1.00 , ~ 1.00
Qwen-7B	+ LoRA	+0.003, +0.000	1.73, 3.74
Qwen-1.5B	gate only	+0.002, -0.002	1.13, 1.13
Qwen-1.5B	+ LoRA	+0.002, -0.002	5.83, 19.9
Llama-8B	gate only	-0.007 , +0.002	1.78, 1.79
Llama-8B	+ LoRA	-0.008 , -0.002	1.5×10^5 , 1.5×10^6
Mistral-7B	gate only	+0.008, -0.002	2.77, 3.03
Mistral-7B	+ LoRA	-0.002 , -0.005	1.8×10^4 , 1.0×10^5

Retrieval-aware warm-up: recognition improves, recall stays at floor

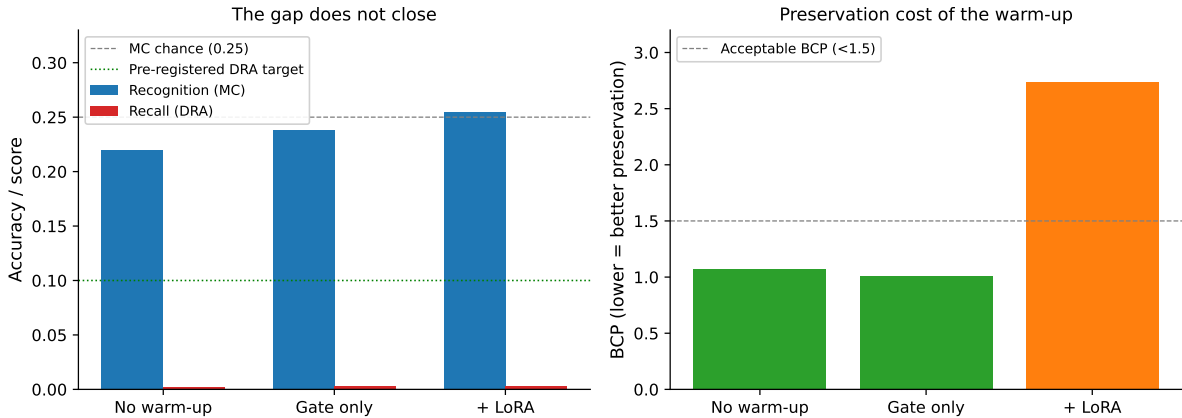


Figure 5: Warm-up extension results. Left: recognition (MC) and recall (DRA) under each condition; the pre-registered DRA target is marked. Recognition rises above chance while recall stays at floor. Right: the preservation cost, showing that the higher-capacity variant buys its small recognition gain with substantial base-capability degradation.

become smooth and seed-stable, while the lottery persists in—and at small scale destroys—the unconstrained baseline.

7.7 Repairing the Gap: A Retrieval-Aware Warm-Up

Proposed: The recognition–recall diagnosis of Section 7.2 implies a specific repair: teach the frozen model to use KV memory during generation before asking it to consolidate. If the gap is a missing *skill* rather than a structural limit, a lightweight warm-up should narrow it.

Setup: We run the warm-up of Section 5.6 on 400 WikiText-2 passages disjoint from the evaluation facts, then re-run the identical recognition and recall evaluation on the 200-fact benchmark with the trained component frozen. We test two capacities: the *gate only* (36 parameters) and *gate + LoRA*, in which W_{cons} trains alongside. Success was pre-registered as $\text{DRA} \in [0.10, 0.15]$ at $\text{BCP} < 1.5$.

Result—two distinct failure modes, replicated across families: The gate-only variant does not train: its loss is flat and the learned scales converge to 0.99–1.00, i.e. the optimizer finds that the identity is

locally optimal. Thirty-six parameters modulating a memory signal cannot manufacture the capability to use it. The gate + LoRA variant *does* learn the objective—on every model the warm-up loss falls substantially (e.g. 4.30 $\rightarrow$ 3.33 on Qwen-7B), confirming genuine acquisition of memory-conditioned prediction—and recognition improves modestly where the model survives (MC 0.220 $\rightarrow$ 0.270 on Qwen-7B, seed 0). But recall does not follow anywhere: across sixteen warm-up runs on four models, $|\Delta\text{DRA}| \leq 0.008$ (Table 7). Meanwhile the LoRA variant’s preservation cost is family-dependent and sometimes catastrophic: mild on Qwen-7B (BCP 1.7–3.7), severe on Qwen-1.5B (5.8–19.9), and destructive on Llama and Mistral (BCP 10^4 – 10^6) at the same untuned learning rate. Two further observations: when LoRA is present the gate remains pinned at exactly 1.000 (gradient flows to the higher-capacity component), and on Mistral the gate-only warm-up substantially *reduced* the interference cost of carrying a populated memory bank (BCP with bank active: 8.9 $\rightarrow$ 2.8)—the gate learns to damp memory it cannot learn to exploit.

Interpretation: The honest statement carries its scope. **A lightweight retrieval-aware warm-up— ≤ 300 steps on a general corpus, with either a minimal gate or a rank-16 adapter—does not close the recognition–recall gap on any of four models spanning three families.** What it does do is deepen the original diagnosis in two ways. First, the gap survived a principled intervention drawn from published precedent [Wu et al., 2022, G-MemLLM Authors, 2026, Wang et al., 2024], replicated across families—stronger evidence of its structural character than any single-model observation. Second, the intervention moved recognition and recall *differentially*—the same training that raised MC accuracy left free-form generation untouched—which is precisely the dissociation the paper names, now reproduced under an experimental manipulation rather than merely observed. We note the caveat that the warm-up learning rate was not tuned per family, which the Llama and Mistral collapses underline; a per-family-tuned variant is cheap follow-up work. We do not claim that retrieval-aware training cannot close the gap; Memorizing Transformers demonstrates that full-scale memory-augmented pretraining does. We claim that the cheap version does not, and we report the scale at which we tested it.

This closes the first movement. Every route into the failure so far—perfect replay data, relaxed safety sweeps, a trained retrieval skill—treated the consolidation target as given. The sections that follow question the target itself.

7.8 Localising the Failure: Substrate and Surface Diversity

Hypothesis: The knowledge-editing literature (Section 2) predicts that factual associations live in mid-stack MLP modules, not the top-third attention projections where biological analogy placed W_{cons} —and that single-wording training should produce string memorization rather than extractable knowledge. If correct, the recall floor of Section 7.4 is a localisation error, not a stability–plasticity limit.

Setup: Five arms isolate the two factors on 50 facts at matched exposure (1,200 training steps), on Qwen2.5-7B and Llama-3.1-8B, two seeds, safety clipping disabled so that placement and data are the only variables. Arms: (i) attention-top placement, single wording (the original); (ii) mid-MLP placement, single wording; (iii) mid-MLP with paraphrase diversity; (iv) context distillation into mid-MLP with paraphrases; (v) context distillation into attention-top (the placement control for the objective). Evaluation is greedy throughout.

Results: Table 8 reports the ladder. Three findings, each consistent across both models and seeds. First, *placement alone halves background damage in 8 of 8 comparisons* (Qwen BCP 3.6 $\rightarrow$ 1.2–1.8 at equal steps) and lifts recall off the floor—the same gradient budget, spent in the substrate the editing literature indicates, does less harm and more good. Second, *paraphrase diversity converts memorization into extraction*: introducing paraphrases moves cloze and free-form recall in opposite directions (0.56 $\rightarrow$ 0.38 cloze, 0.03 $\rightarrow$ 0.22 DRA), which is precisely the signature of the model ceasing to reproduce a token sequence and beginning to store a retrievable association. Third, *distillation from the in-context teacher is the gentlest write*: the best arm reaches DRA = 0.36 at BCP = 0.83 on Qwen—sixty times the original floor, at *better-than-baseline* preservation—while the placement control (same objective, original substrate) recovers almost none of it. On Llama the best arm is paraphrase-CE at 0.353. The recall floor of the first movement was therefore not

Table 8: Five-arm localisation ladder (50 facts, matched exposure, clipping disabled; Qwen2.5-7B shown, seed means). Placement, diversity, and objective stack. Cloze and free-form move in *opposite* directions when paraphrases are introduced: the model stops memorizing the string and starts knowing the fact.

Arm	Free-form DRA	Cloze	BCP
attention-top, single wording (original)	0.01	0.44	3.6
mid-MLP, single wording	0.03	0.56	1.2–1.8
mid-MLP + paraphrases	0.22	0.38	1.4
mid-MLP + paraphrases + distillation	0.36	0.31	0.83
attention-top + distillation (control)	0.06	0.35	2.9

a structural property of frozen-backbone consolidation. It was a map error: the pipeline was writing to the wrong address, in the wrong format.

7.9 The Third Cause: The Safety Machinery Erases the Write

Setup: The ladder ran with clipping disabled. Wiring the winning recipe into the *full* autonomous pipeline—tagging, PRP selection, sleep engine, safety machinery, two-stage validation—tests whether it survives the system it must live in. We run the complete pipeline on 200 facts with the safety machinery at its designed setting ($\delta_{\max} = 0.01$, plasticity scaling on), and identically with clip and scaling relaxed, three seeds each on Qwen2.5-7B.

Results: As designed, the machinery nullifies the recipe: trained-subset DRA 0.007/0.007/0.007, zero consolidations confirmed by the direct recall gate—indistinguishable from the original floor. Relaxed, the identical pipeline reaches trained-subset DRA **0.210 $\pm$ 0.018** at BCP = 0.90–0.99, with 21–24 of $\sim$ 50 selected facts passing the external greedy recall gate: the first externally verified autonomous consolidations in the programme. The original consolidation failure therefore had *three* stacked causes: wrong substrate (§7.8), wrong surface statistics (*ibid.*), and a safety clip calibrated so tightly that it erased the write after training performed it—consistent with the $\delta_{\max}$ saturation geometry identified in the diagnosis era. Each cause alone suffices to hold recall at floor, which is why no single-factor intervention in the first movement could move it.

7.10 The Repaired Pipeline as an Autonomous System

The findings above fix the revised operating point (Section 5.7): mid-MLP W_{cons} , paraphrase replay, context distillation, moderate clipping ($\delta_{\max} = 0.1$, scaling on), two-stage validation with the direct recall gate, greedy evaluation. At this operating point the full pipeline—wake-phase tagging and KV writes, PRP selection, autonomous sleep, external validation—achieves single-cycle trained-subset DRA of 0.189 ± 0.031 at BCP = 0.882 ± 0.027 on Qwen2.5-7B across five seeds: recall statistically indistinguishable from the relaxed setting, at measurably *better* preservation, with every seed below BCP = 1.0. On the frontier of Figure 1, whose best original point bought DRA = 0.103 at BCP = 2.73, this configuration sits far above the upper-left intersection that no diagnosis-era setting could reach. The two-stage gate confirms 16–23 consolidations per run externally—against the diagnosis era’s 0.4 ± 0.5 .

7.11 The Validation Matrix: Three Families, Two Scales, Two Baselines

All results in this section were run under pre-registered gates (Section 6); predictions were committed before execution.

Single-cycle generalisation (5 seeds per model): Table 9 reports the repaired pipeline across the four models. Per-model recipes were fixed by a pre-registered tuning gate (best of distillation vs. paraphrase-CE at trained-subset DRA ≥ 0.10 , BCP ≤ 1.5 , seed 0): distillation won everywhere it was tried (on Mistral, 0.750 @ 1.31 vs. 0.655 @ 2.98—the in-context-teacher write is also the gentler write). Every model exceeds

Table 9: The repaired pipeline across four models: single-cycle trained-subset DRA and BCP (5 seeds, mean $\pm$ SD). Recipes fixed by pre-registered tuning gates. Every model is $\geq 17\times$ the diagnosis-era floor of ≈ 0.006 ; the strongest result is on the family the recipe never saw during development.

Model	Recipe	Trained-subset DRA	BCP
Mistral-7B	distill, $\delta_{\max}=0.1$	0.750 ± 0.033	1.205 ± 0.058
Llama-3.1-8B	par.-CE, $\delta_{\max}=0.03$	0.512 ± 0.048	2.088 ± 0.140
Qwen2.5-7B	distill, $\delta_{\max}=0.1$	0.189 ± 0.031	0.882 ± 0.027
Qwen2.5-1.5B	distill, $\delta_{\max}=0.1$	0.105 ± 0.020	1.043 ± 0.016

Table 10: Ten-cycle validation: cumulative DRA at cycle 10 (seeds 0/1) with BCP. The repaired pipeline exceeds naive on cumulative recall on all four models, both seeds; at 1.5B the unconstrained arm’s damage blows up seed-randomly while the constrained arm is stable.

Model	SLEEP v2-moderate	Naive LoRA
Qwen2.5-7B	$0.173/0.175$ @ 1.87/2.24	$0.053/0.060$ @ 1.66/1.80
Mistral-7B	$0.173/0.168$ @ 1.18/1.18	$0.118/0.160$ @ 1.71/1.57
Qwen2.5-1.5B	$0.097/0.102$ @ 2.35/2.49	$0.038/0.050$ @ $24.5/11.4$
Llama-3.1-8B	$0.102/0.062$ @ 4.33/9.20	$0.082/0.042$ @ 2.24/1.67

the diagnosis-era floor by $17\times$ or more. The effect is *strongest off the development model*—Mistral-7B, which the recipe had never touched, is the best result in the programme—which argues against the recipe being overfit to Qwen. The 1.5B result (0.105, thin but gate-passing, only 10 of 42 selected facts surviving the recall gate versus 29 of 31 on Mistral-7B) suggests extraction capacity grows with scale. Llama-3.1-8B is the flagged exception: a pre-registered clip ladder ($\delta_{\max} \in \{0.1, 0.05, 0.03\}$) reduced damage monotonically ($4.35 \rightarrow 3.64 \rightarrow 2.14$) with recall degrading only slowly ($0.659 \rightarrow 0.593 \rightarrow 0.520$), but never reached the 1.5 bar—Llama’s damage is *not clip-dominated*, and per-family learning-rate tuning is the named open item. It runs the matrix at $\delta_{\max} = 0.03$ and is reported as *not yet tuned*, not as solved.

Ten cycles: the design-intent claim, tested (2 seeds per arm): Table 10 repeats the ten-cycle protocol of Section 7.6 with the repaired pipeline (v2-moderate) against naive LoRA at the same mid-MLP adapter geometry. The pre-registered prediction—v2 exceeds naive on cumulative recall on at least two of the three newly-tested models, both seeds—was met on three of three; with Qwen2.5-7B it holds on all four models. Three regularities are seed-consistent. First, **the design-intent claim is now real**: the safety-constrained system accumulates more recall than unconstrained fine-tuning over ten cycles, $3\times$ on the anchor model. Second, **the machinery is load-bearing at small scale**: naive LoRA detonates Qwen2.5-1.5B (BCP = 24.5 and 11.4; the blow-up size is seed-random—the plateau lottery of Section 7.6, reproduced), while the constrained arm holds ≈ 2.4 on both seeds. Third, **the constrained arm no longer exhibits the lottery**: its across-seed spread is ≈ 0.005 DRA on every model against naive’s 0.04–0.06, and its damage drifts smoothly with no step-cliffs. The seed-instability finding of Section 7.6 thereby resolves into a property of *unconstrained* arms specifically. Llama repeats its single-cycle pattern—more recall than naive on both seeds, at damage that compounds unacceptably (BCP 4.3/9.2)—and keeps its flag. One qualitative signature recurs on every model: naive scores higher on *current-cycle* recall while the constrained arm retains *earlier* cycles better—memorization versus consolidation, now visible in the horizon dimension.

The baseline that matters: EWC-only (matched arms): The diagnosis era left an uncomfortable finding: EWC alone looked competitive with the full pipeline. The matrix settles it with the matched-arm comparison of Section 6 on the two headline families, ten cycles, two seeds (Table 11). The pre-registered prediction held exactly: the penalty gives EWC-only the best damage numbers, but its recall sits at naive’s level—**regularization prevents forgetting; it does not create extraction**. The repaired pipeline delivers $3\times$ the cumulative recall of both baselines on Qwen2.5-7B at comparable damage, passing the pre-

Table 11: The EWC showdown: cumulative DRA at cycle 10 (seeds 0/1) with BCP, matched-arm EWC. On the anchor model the repaired pipeline triples both baselines’ recall at comparable damage; EWC’s penalty preserves but does not create recall.

Model	SLEEP v2-moderate	EWC-only	Naive LoRA
Qwen2.5-7B	0.173/0.175 @ 1.87/2.24	0.060/0.052 @ 1.53/1.80	0.053/0.060 @ 1.66/1.80
Llama-3.1-8B	0.102/0.062 @ 4.33/9.20	0.075/0.072 @ 2.32/1.79	0.082/0.042 @ 2.24/1.67

registered headline gate. On Llama, EWC-only is competitive with the not-yet-tuned v2 (0.075/0.072 at far better BCP)—reported plainly: the Llama column is not a v2 win.

Residual gaps, named: Three limitations survive the matrix and are carried explicitly into future work. Background damage leaks $\approx +0.12$ BCP per cycle in every arm—no configuration holds $\text{BCP} < 1.05$ over ten cycles. Early-cycle batches fade in cumulative recall because the engine rehearses only the current batch; interleaved rehearsal of prior consolidations—the mechanism biological sleep actually implements—is the clearest next component. And Llama-3.1-8B requires per-family optimisation before its recall gains are usable.

8 Discussion

Theoretical contribution: The paper’s central theoretical result is a causal decomposition of why biologically-faithful consolidation failed—and the demonstration that each cause, once named, is removable. The recall floor that survived every within-paradigm intervention (perfect replay data, safety sweeps, a trained retrieval skill) decomposed into three stacked errors: substrate (the adapter sat in attention projections chosen by anatomical analogy, not the mid-stack MLPs where causal tracing places factual associations), surface statistics (single-wording training produces string memorization, visible as cloze success with recall failure—and paraphrase diversity moves the two metrics in *opposite* directions), and calibration (a hard clip tight enough to erase the write after training performed it). This carries a general lesson for biologically-inspired machine learning: biological *mechanisms* can transfer while biological *anatomy* does not. Tagging, resource allocation, replay, and sleep-phase consolidation all survived the transfer; the hippocampus-to-neocortex layer mapping was the part that failed, and the transformer’s own “anatomy”—as revealed by the editing literature [Geva et al., 2021, Meng et al., 2022, 2023]—had to replace it. The diagnostic findings stand independently: the recognition–recall dissociation for attention-injected memory (now reproduced under an experimental manipulation via the warm-up), the near-total failure of proxy self-validation (93–100% false confirmation on every model), and the horizon-and-seed instability of unconstrained continual-learning baselines, which the validation matrix sharpens into an asymmetric claim: constrained arms can be made seed-stable; unconstrained arms are the lottery.

The EWC dissociation: The matched-arm showdown isolates a distinction the continual-learning literature tends to blur: *preservation* and *acquisition* are different capabilities. EWC-only achieves the best background-damage numbers in the matrix while its recall never rises above naive fine-tuning’s—the Fisher penalty determines what the model *keeps*, not what it can *extract*. The repaired pipeline’s advantage comes from the write path (where and on what surface statistics the gradient lands), not from protecting the result harder. Systems that report only forgetting metrics would score EWC-only the winner of an experiment it loses three-fold on recall.

Practical implications: For practitioners, the operational lessons now come in two tiers. From the diagnosis: direct-write KV memory supports recognition-style tasks on a frozen backbone but not free-form generation, and a cheap retrieval-aware warm-up does not change that; proxy self-validation must be audited against an external criterion before it is trusted; and continual-learning comparisons involving unconstrained arms need horizon sweeps and seed spreads, because a single run’s ordering is lottery-decided. From the repair: consolidation adapters for factual content belong in mid-stack MLPs, not attention; train on paraphrase

diversity including question–answer forms, or the system will memorize strings it cannot use; distillation from the model’s own in-context behaviour is both the strongest and the gentlest write we measured; and safety machinery should be calibrated by ablation in its design regime—ours was simultaneously too tight (erasing writes single-cycle) and, once moderated, genuinely load-bearing (preventing a $24\times$ damage blow-up at 1.5B that unconstrained fine-tuning incurs).

Challenges: Several scope constraints bound the strength of our claims:

- (i) We test four models across three families, all dense decoder-only architectures at $\leq 8\text{B}$; mixture-of-experts and much larger scales remain untested.
- (ii) We test *synthetic* facts in 10 templates with template-generated paraphrases; natural document content, and naturally-occurring paraphrase distributions, may differ.
- (iii) Llama-3.1-8B is validated for recall gains but not for preservation: no tested clip reaches our damage bar, its multi-cycle BCP compounds (4.3–9.2), and EWC-only is competitive with the repaired pipeline on that family. We flag it as *not yet tuned* and treat per-family learning-rate optimisation as required, not optional.
- (iv) We force *eager attention*; SDPA and FlashAttention paths handle 4D additive masks differently and we did not characterize them after the gating mechanism was finalized.
- (v) No configuration holds $\text{BCP} < 1.05$ over ten cycles—damage leaks $\approx +0.12/\text{cycle}$ in every arm—and cumulative recall decays as early batches fade, because the engine rehearses only the current batch. The validated system is the best of its cohort, not a solved problem.
- (vi) The multi-cycle validation uses two seeds per arm (with five seeds single-cycle); the diagnosis-era lottery finding cautions against over-reading two-seed orderings, though every multi-cycle claim we make held on both seeds and the constrained arm’s variance is demonstrably small.

Where variance was measured it now separates into three regimes. The diagnostic floors are tight (DRA $\text{SD} \leq 0.003$). The repaired pipeline is tight (single-cycle $\text{SD} \leq 0.05$ on every model; multi-cycle across-seed spread ≈ 0.005). The unconstrained baseline is not (multi-cycle damage varying up to $24\times$ seed-to-seed at 1.5B)—and we report that asymmetry as a finding. The strongest claim our evidence supports is: on these models, on this dataset, under pre-registered gates, the repaired pipeline’s advantages replicate on every seed we ran.

Ethical considerations: Continual-learning systems for language models, if they work, would reduce re-training costs substantially and enable models to learn from interaction. The repair reported here moves that prospect closer, which sharpens rather than softens the concerns: persistent memory of user inputs, model memorization of sensitive information, and asymmetric power between systems that can update themselves and users who cannot. Our architecture stores user inputs (tagged spans) in a KV bank and consolidates a subset into permanent weights; deployed naively, this would amount to silent on-device data retention—now with a consolidation path that demonstrably works. We argue that any deployed continual-learning system must include explicit user controls over what is encoded, what is consolidated, and what is permanent, alongside mechanisms for selective forgetting on user request. The self-grading result adds a safety dimension: a system that certifies its own learning against an internal proxy may report successful consolidation of information it cannot reproduce—or fail to report retention of information it can. External validation gates of the kind we adopt should be considered a safety component, not merely a metrological one.

9 Conclusion and Future Scope

We implemented a biologically-faithful memory consolidation architecture for transformer language models, diagnosed its failure to the point of causal decomposition, and repaired it. The diagnosis stands on four models from three families, five seeds each: direct-write KV memory produces recognition without recall (a floor of $\text{DRA} \approx 0.006$ from models that follow the same facts in context at up to 0.99); a retrieval-aware warm-up moves recognition but never recall; internal surprise-based validation false-confirms 93–100% of

consolidations; and unconstrained continual-learning baselines are horizon- and seed-unstable. The repair stands on the same models under pre-registered gates: relocating consolidation to mid-stack MLPs, training on paraphrase diversity, distilling from an in-context teacher, and moderating the safety clip raises single-cycle recall to 17–125 $\times$ the floor (0.750 ± 0.033 on Mistral-7B), beats naive fine-tuning on cumulative ten-cycle recall on all four models on both seeds, beats a matched EWC-only baseline three-fold on the anchor model, and prevents the seed-random damage catastrophe that destroys unconstrained fine-tuning at small scale. The central lesson generalises beyond this system: in transferring biological memory to transformers, the *mechanisms* (tagging, selective replay, staged consolidation, sleep-phase scheduling) survive; the *anatomy* (which layers play hippocampus and neocortex) must come from the transformer’s own causal structure, and the knowledge-editing literature already maps it.

Future scope: Our results give follow-up work specific targets rather than vague directions:

- (i) *Interleaved rehearsal*: cumulative recall decays because the engine rehearses only the current batch; biological sleep interleaves old consolidations with new. Adding rehearsal of prior cycles’ facts (or their paraphrases) to the replay mix is the clearest next mechanism and directly targets the early-batch fade.
- (ii) *Per-family optimisation*: Llama-3.1-8B gains recall like every other model but bleeds damage that no clip setting contains; its learning-rate and signal-calibration surface is unexplored. The matrix provides the harness; the sweep is cheap.
- (iii) *Closing the damage leak*: every arm leaks $\approx +0.12$ BCP per cycle. Candidate mechanisms: per-cycle Fisher refresh against the *original* base model rather than the drifting anchor, and consolidation-aware early stopping per fact.
- (iv) *The recognition–recall gap at training scale*: our warm-up failed at ≤ 300 steps; the informative next experiment remains the same intervention at Memorizing-Transformers scale with per-family learning rates, now with the mid-MLP consolidation path available as the comparison arm.
- (v) *Beyond dense $\leq 8B$ decoders*: extend the matrix to mixture-of-experts and larger scales, where the editing literature’s localisation maps—and therefore the substrate correction—may differ.

We release the implementation, formalization, pre-registrations, full experimental logs, and all validation-phase runs to enable that work.

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Annexure A Formalization Amendments

We resolved 36 design questions in a pre-registered formalization document before any code was written. Empirical work surfaced five amendments, each documented with the experiment that motivated it. We summarize them here; the full formalization (including unchanged questions) is released alongside the implementation.

A.1 ($\delta_{\max}$ must scale with model size): The original specification of $\delta_{\max} = 0.001$ as a per-parameter clip bound saturates within ~ 30 training steps on a 7B model with rank-16 LoRA: 36/36 LoRA parameters hit the clip on every step thereafter, and validation always fails. Required correction: $\delta_{\max} \approx 0.01$ at 7B scale. The per-parameter bound is the wrong abstraction; what matters is the Frobenius norm of the LoRA update relative to the model’s natural weight scale. A scaling rule of $\delta_{\max}(d) = 0.001 \cdot \sqrt{d/768}$ partially corrects this but underestimates the empirically required value.

A.2 (α_{slow} fails for two separable reasons): The original specification of $\alpha_{\text{slow}} = 10^{-5}$ produces gradient updates that round to zero in bfloat16 storage: training loss is completely flat across 100 steps. Bumping to 10^{-4} produces a phase transition (loss $1.46 \rightarrow 0.55$). We subsequently isolated whether this is purely a dtype artefact by re-running the original default in float32 on the same benchmark. It is not. In float32 the loss does move—but only from 2.902 to 2.874 over 100 steps ($\sim 1\%$), with 0/12 validation passes and BCP = 0.998, i.e. the adapter is effectively unmoved. There are therefore *two stacked failures*: a genuine bfloat16 precision floor (updates round to exact zero) *and* a learning rate that is under-scaled for a 7B model in any precision. Required correction: α_{slow} must clear both bounds— $\alpha_{\text{slow}} \geq 10 \cdot \text{ulp}(\text{dtype})$ for representability, and a magnitude that scales with model size for effect. The original 10^{-5} implicitly assumed both float32 storage and a smaller model.

A.3 (μ_{surprise} has a generator-discriminator gap): The quality check “replay must be at least as surprising as the model’s baseline” rejects all replays in practice, because replays are generated from the same model that defines the surprise function and are therefore systematically more in-distribution than external calibration text. The empirical surprise of generated replays (~ 1.25 nats) falls below the baseline computed on calibration text (~ 1.41 nats). Required correction: replace the absolute threshold with a relative-gain criterion ($\text{surprise}(\text{replay}) > \beta \cdot \text{surprise}(\text{seed})$) or compute the threshold from self-generated samples. Until amended, we disable the absolute check.

A.4 (Validation has a data-volume floor): The validation criterion $\text{surprise}_{\text{new}} < (1 - \epsilon_{\text{learn}}) \cdot \text{surprise}_{\text{old}}$ with $\epsilon_{\text{learn}} = 0.10$ is achievable only above a minimum number of replays. With 5 facts and 2–3 replays, 0% of candidates pass; with 200 facts and ~ 50 replays, 40%–60% pass depending on safety settings. The criterion itself is sound, but the formalization should explicitly document that pass rates are zero below a benchmark-dependent volume threshold.

A.5 (Tags are pointers, not storage): The original formalization stored only the K/V projections of tagged tokens, but tagged tokens are short novel-entity mentions (4–8 tokens) that capture the location of the surprise without its content. Required correction: when a tag fires, store K/V for the full sentence containing the tagged span (the *episode*), not just the tagged tokens. This corresponds to the proposal’s own framing of tags as pointers but was implemented as direct storage. Section 7.3 reports the empirical correction.

Annexure B Hyperparameter Table

Annexure C Implementation Details

C.1 KV Memory Injection Mechanics

KV memory injection requires care to avoid silently wrong attention scores. We document the choices made:

Table 12: Full hyperparameter list. “Source” indicates which of the 36 formalization questions specifies the parameter. “Status” indicates whether the value matches the original specification or was empirically amended.

Parameter	Value	Source	Status
Tagging κ (z-score threshold)	1.5	Q1.2	default
Tagging g (gap tolerance)	3	Q1.3	default
Tagging min span length	4	Q1.3	default
Tagging EMA β	0.99	Q1.2	default
PRP weights (w_E, w_a, w_x, w_r)	(0.35, 0.30, 0.15, 0.20)	Q2.2	default
PRP budget multiplier c_{prp}	500	Q2.4	default
PRP cross-ref similarity threshold	0.5	Q2.2	default
LoRA rank (r)	16	Q3.1	default
LoRA α	32	Q3.1	default
Adapted layer fraction	0.333 (top 1/3)	Q3.2	default
Adapted matrices	V, O	Q3.1	default
α_{fast}	10^{-4}	Q3.3	default (LoRA W_{fast} only)
α_{slow}	10^{-4}	Q3.3	amended (A.2)
δ_{max}	0.01	Q3.4	amended (A.1)
λ_{EWC}	100	Q3.4	default
ϕ_{min} (plasticity floor)	0.10	Q3.4	default
ϵ_{learn} (validation)	0.10	Q4.6	default
$\epsilon_{\text{degrade}}$ (rollback)	0.02	Q4.5	default
Replay temperature	0.7	Q4.1	default
Replay top- p	0.9	Q4.1	default
Compression target	5	Q4.1	default
Steps per memory	5	Q4.4	default
Sleep weight decay	0.01	Q4.4	default
KV bank capacity (tokens)	20,000	—	new (A.5)
KV top- k retrieval gate	16 (default), 64 (episode)	—	new (A.5)
KV memory storage unit	full sentence (episode)	—	amended (A.5)
μ_{surprise} (quality check)	disabled (set to 0)	Q4.1	amended (A.3)

Position encoding. We treat memory entries as occupying negative position IDs $[-n_{\text{mem}}, -1]$. RoPE [Su et al., 2024] is applied at read time to memory keys with these positions. The encoding is mathematically valid (sin is odd, cos is even, both well-defined at negative inputs) and verified unit-tested. Memory thus appears to current queries (at positive positions) as “recent context tokens,” a pattern the model was trained on through RoPE’s relative-position mechanic.

Attention mask. Qwen2 uses float additive masks with $-\infty$ for masked positions and 0.0 for visible. When extending the mask with memory columns, we fill those columns with 0.0 of the mask’s own dtype. A boolean True or integer 1 here would silently produce wrong scores on float-mask attention paths.

Cache ordering. Memory K/V must be injected *after* the `past_key_values` cache update, not before. If memory is injected first, the cache accumulates a duplicate copy of memory tokens on every autoregressive generation step; by step 50 the cache contains 50 copies of memory and BCP collapses (we observed $\sim 96\times$ degradation under this bug).

K/V extraction layer-norm. When extracting K/V from a forward pass via `output_hidden_states=True`, the apparent “hidden state at layer ℓ ” is the input to layer ℓ , but attention computes K/V from the layer-normed input. Skipping the layer-norm produces silently wrong K/V. We add an integration test that captures K/V via a forward hook during a normal forward pass and asserts equality with our extraction.

C.2 Per-Phase KV State

A correctness requirement we discovered: KV injection must be *disabled* during W_{cons} training. If left enabled, the model can satisfy the training loss by relying on injected memory rather than encoding the replay content into weights—a crutch the model would not have at inference time after the bank is cleared. We add a `set_enabled(bool)` method to the injector and toggle as follows:

- Wake (tagging, KV write): enabled.
- Phase 1 Generate (replay): enabled (so the model attends to source while compressing).
- Phase 2 Quality check: state preserved (no parameter changes; result invariant).
- Phase 3 Train (W_{cons} updates): *disabled* (forces W_{cons} to encode into weights).
- Phase 4 Perplexity check: disabled (clean preservation measurement).
- Phase 5 Validate & cleanup: disabled (clean recall measurement).
- After successful consolidation: bank is cleared, enabled flag reset for next cycle.

C.3 Memory Gate Placement

The warm-up’s memory gate (Section 5.6) is applied to the memory *values* immediately before they are concatenated onto the key/value time axis, and after the memory keys have received their negative-position RoPE. Placement matters for a reason worth recording: attention output is linear in V but non-linear in K , so gating values scales exactly the memory contribution to the residual stream, whereas gating keys would perturb the softmax allocation itself. A consequence of this choice is that the gate cannot reclaim attention mass—memory positions still absorb softmax weight even at $g \rightarrow 0$ —so it modulates the strength of the memory contribution rather than switching it off. Because the gate is initialised to the identity, installing it is a provable no-op: our first test asserts bit-identical logits against the un-gated path, which caught the alternative placements during development.

Annexure D Multi-Cycle Configuration

The multi-cycle experiment of Section 7.6 uses Setting A from the Pareto sweep ($\delta_{\text{max}} = 0.02$, $\lambda_{\text{EWC}} = 50$, $\alpha_{\text{slow}} = 10^{-4}$, steps-per-memory = 4) for the SLEEP arm. The naive LoRA arm uses 200 gradient steps per cycle on the cycle’s batch only, with AdamW (lr 10^{-4} , weight decay 0.01, batch size 32, uniform-with-replacement sampling). Both arms accumulate adapter state across cycles; weights from cycle i persist into cycle $i + 1$. Both arms are evaluated on identical control texts and recall benchmarks after each cycle. Replay strategy for the SLEEP arm is set to `original` (replays are the original fact tokens), eliminating replay quality as a confound and isolating the multi-cycle dynamics of the consolidation training step. The ten-cycle programme divides the 200 facts into 10 disjoint batches of 20 and repeats the paired protocol with three seeds on Qwen2.5-7B and two seeds on each of Qwen2.5-1.5B, Llama-3.1-8B, and Mistral-7B (nine run-pairs). The earlier three-cycle configuration used batches of ≈ 67 , and we note that batch size per cycle is a candidate moderator of the plateau onset.

Annexure E Validation-Matrix Protocol

The revised-pipeline validation of Section 7.11 was executed as a staged campaign whose gates, predictions, and stop-loss were committed to the repository before any run: (D0) the three-check ground-truth verification gate on all four mid-MLP configurations; (D1) per-family recipe tuning at seed 0, advancement criterion trained-subset DRA ≥ 0.10 at BCP ≤ 1.5 , plus a mid-campaign-registered clip ladder for Llama with its selection rule fixed before execution; (D2) single-cycle, five seeds per model at the tuned recipes (200 facts, 1,500 training steps); (D3) ten-cycle, both arms, two seeds per model (20 facts/cycle, 400 distillation steps/cycle for the SLEEP arm, matched mid-MLP adapter geometry for the naive arm); (D4) the matched-arm EWC-only comparison (online EWC, $\lambda = 100$, Fisher accumulated over each cycle’s facts and re-anchored after consolidation, at most 20 Fisher samples per cycle). The v2-moderate operating point

throughout is $\delta_{\max} = 0.1$ (0.03 for Llama per its ladder), plasticity scaling on, $\lambda_{\text{EWC}} = 0$ within the sleep-train step, $\alpha_{\text{slow}} = 10^{-4}$. The campaign comprised 35 runs on a single RTX A6000 in one 19.7-hour queue with zero failures; pre-registration documents, per-run JSON results, and execution logs are released in the repository (`logbook/` and `experiments/results/`).

Annexure F Reproducibility Notes

Seeding. A single entry point seeds Python’s `random`, NumPy, PyTorch (CPU and CUDA), and `PYTHONHASHSEED`. Multi-seed experiments are executed one seed per process so that model initialisation is genuinely independent rather than sequentially derived; the aggregating harness collects per-run JSON and reports mean and sample standard deviation.

Environment sensitivity. Three environment issues materially affected results and are recorded so that replication does not rediscover them. First, `transformers` must be pinned to the version documented in the repository: a later release imports a distributed-tensor symbol unavailable in the CUDA image’s PyTorch build, and fails at import. Second, adapter modules must be constructed in the wrapped layer’s dtype; a float32 adapter attached to a bfloat16 model raises a dtype mismatch at the first matrix multiply. Third, the patched attention forward must not assume architecture-specific attributes: `sliding_window` exists on Qwen2 attention modules but not on Llama’s or (in the pinned release) Mistral’s, and is read defensively. The cross-architecture verification harness of Section 6 caught the third issue before any invalid run executed; all are recorded in the released logs.

Verification practice. Every modification to the KV-injection path is accompanied by a hook-based equality test against the model’s own computation, written before the feature it verifies. This practice caught two silent-failure bugs during development: a missing layer-norm in K/V extraction, and a cache-ordering error that duplicated memory on every generation step. The memory gate’s identity-initialisation test follows the same pattern.